



KOLFRAM EDUCATIONAL SERVICES KAMPALA
PRE NATIONAL MOCK SET 1 EXAMINATIONS
2026

MATHEMATICS

Time Allowed: 2 Hours 30 Minutes



Random No.						Personal No.		

Candidate's Name:

Candidate's Signature:.....

School Name:

Read and follow these instructions carefully:

1. This paper has **two** sections: **A** and **B**. Section **A** has **20** questions and section **B** has **12** questions. The paper has **12** printed pages.
2. Answer **all** questions. **All** answers to both sections A and B must be shown in the spaces provided.
3. All answers **must** be written using a **blue** or **black** ball point pen or ink. Any answer written in pencils other than on graphs and diagrams will **not** be marked.
4. No calculators or **electronic** pens are allowed in the examination room.
5. Unnecessary **changes** in your work and handwriting that cannot be read easily may lead to **loss of marks**.
6. Do not fill anything in the table indicated: "**FOR EXAMINERS' USE ONLY**" and boxes inside the question paper.

FOR EXAMINER'S USE ONLY		
QN. NO.	MARKS	EX'ER'S INITIAL
1-5		
6-10		
11-15		
16-20		
21-22		
23-24		
25-26		
27-28		
29-30		
31-32		
TOTAL		

SECTION A (40 MARKS)

(Questions 1 to 20 carry 2 marks each)

1. Add $43 + 63$

2. Given that $\mathbf{K} = \{1, 2, 3, 4, 5\}$, find the number of subsets in set $\mathbf{K}$.

3. Round off 54.67 to the nearest whole number.

4. Find the next number in the sequence below
2, 5, 8, 11, 14, _____

5. Simplify: $-2 - -8$

6. Find the value of m in: $5(2m + 1) = 10$.

7. A car uses 7 litres of petrol for every 25km. How many litres are needed for a journey of 75km?

8. Express 00:25 hours in 24-hour clock system.

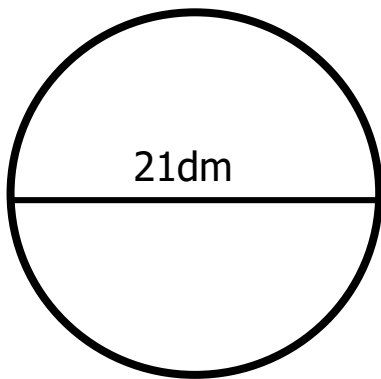
9. A shopkeeper bought 50kg bag at sh.165,000. At what price must he sell the sugar in order to make a profit of sh. 25,000?

10. Find the value of m if: $22m = 122_{\text{three}}$

11. The bearing of A from B is 250° . What is the bearing of B from A?

12. The total surface area of a cube is 24cm^2 . Find its volume.

13. Find the area of a semi-circle whose radius is 21 dm. (Take $\pi = \frac{22}{7}$)



14. In a class, the ratio of boys to girls is 3:2. If there are 15 more boys than girls, how many pupils are in the class?

15. Kigo bus covers a distance of 280km in 4 hours. Find the distance it will cover in 150 minutes at the same speed.

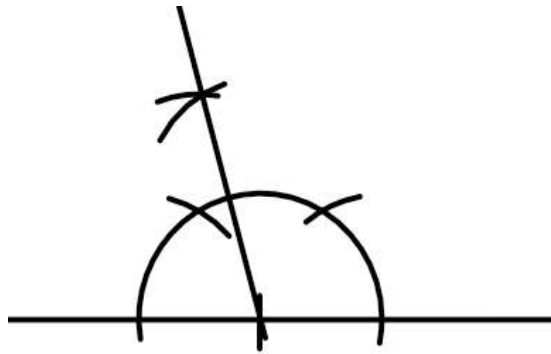
16. Write 0.2333...as fraction in its lowest terms.

17. Maria withdrew consecutive notes of Sh. 20,000 ranging from AB121600 to AB121700. How much did she withdraw?

18. Express 0.00457 in scientific notation.

19. Ruth scored the following marks in a series of tests: 63, 48, 51, 73, 66 and 51. What was his median mark?

20. Using a protractor, to measure the angle constructed below.



SECTION B (60 MARKS)

21. James is 8 yrs older than his friend Peter. The average age of his other 4 friends is 56 years. If Peter and James join the group, the average age becomes 52 years.

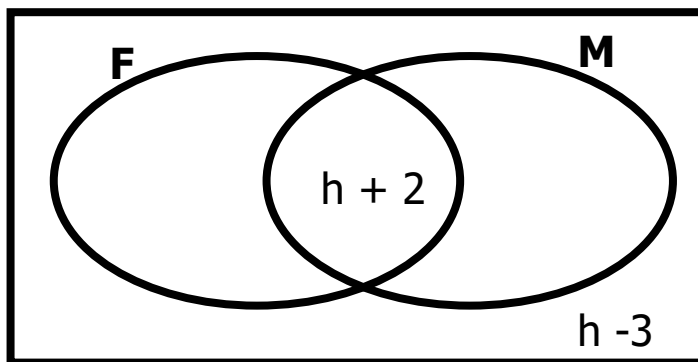
(a) Find James' age.

(b) How old is Peter?

22. 35 guests were invited to a certain party, 22 of the liked meat (M), $h+2$ guests liked both the fish (F) and meat (M), $(h-3)$ ate none of these.

a) Complete the Venn diagram below.

$$\cap (\Sigma) = 35$$



b) Find the value of h

c) How many guests ate fish?

23. Musolini used his salary as follows: $\frac{1}{3}$ on fees, $\frac{1}{4}$ on transport, $\frac{1}{8}$ on clothes and saved the rest. He saved sh. 280,000. Find the fraction saved. *(3 marks)*

(b) How much was his salary? *(2 marks)*

24. (a) Solve: $0.8(h-2) - 0.25(h-6) = 1$

- (b) Okello is 18 years old while Adong is 4 years old. In how many years' time will Adong be half as old as Okello?

25. (a) Using a ruler and a pair of compasses only, construct a triangle XYZ in which $XY = 7.4\text{cm}$ and $\angle XYZ = 30^\circ$ and angle $XYZ = 75^\circ$

(b) Measure YZ

YZ = _____ cm

26. Jacob went shopping and bought the following items:

2 loaves of bread at sh. 5,500 per loaf

3 bars of soap for sh. 6,200 per bar

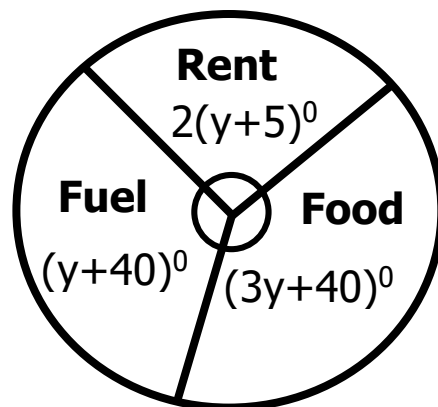
$2\frac{1}{2}$ kgs of salt for sh. 600 per 500kg

(a) What was his total expenditure?

(b) If he was given 10% discount, how much did he pay for all the items?

27. A tank is exposed to three taps: tap A can fill the tank in 8 minutes, tank B can fill the same tank in 6 minutes while tap C can empty the tank in 4 minutes. How much time will the taps take to fill the tank if all the three taps are open at the same time? *(5 marks)*

28. The pie-chart below shows how Mark spends his salary. Study it and answer the questions that follow.



- a) Find the angle sector for food.

(2 marks)

b) If he spends UGX 170,000 on others, how much does he spend on Rent? **(3 marks)**

29. (a) A rectangular polygon has one of its exterior angle of 40° .
How many sides has the polygon? **(2 marks)**

(b) How many right angles does the polygon have? **(2 marks)**

(c) What is the interior angle sum of the polygon? **(2 marks)**

30. Ali, Kato and Lawrence contributed some money to start a business in the ratio 2:5:3 respectively, if Kato contributed UGX 450,000.

(a) How much money did they contribute altogether? **(2 marks)**

(b) Write the amount contributed by Lawrence in words. **(2 marks)**

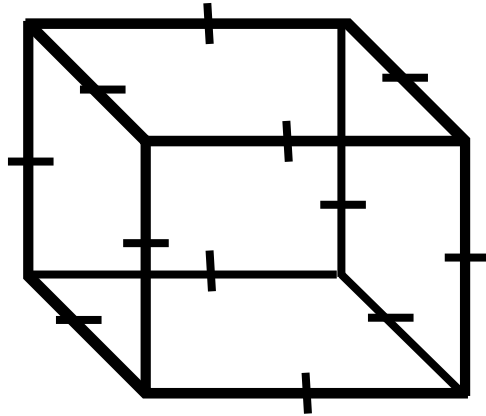
31. (a) Using a ruler, pencil and a pair of compasses only, construct a triangle ABC in which $AB = 8\text{cm}$, $BC = 6\text{cm}$ and $\angle ABC = 120^\circ$.

(4 marks)

(b) Measure length AC

(2 marks)

32. Tendo made a cubic frame for his toy car from a wire of length 960cm.



(a) How long is each side of the cube?

(2 marks)

(b) Find the total Surface area of the frame of the toy car.

(2 marks)

(c) Find the capacity of the cube of the frame of the toy car made.

(2 marks)